

Profile

MSc in **AI & Robotics** with BSc in **Aerospace Engineering**. Specializing in **vision-guided robotic automation** for agricultural applications. Current thesis developing autonomous grapevine pruning systems with real-world deployment focus. Expertise in **ROS**, **computer vision pipelines**, **robotic manipulation**, and production-oriented R&D. Strong background in fluid dynamics, propulsion systems, and control theory.

Professional Experience

- Nov 2025 – Present **Robotics & Computer Vision Engineer (Thesis)**, *FieldRobotics*, Bologna
- Developing **autonomous grapevine pruning system** for commercial agricultural rover deployment.
 - Working on integration of **ABB robotic arm** with ROS1 vision pipeline using native ABB software, ROS, and custom interfaces.
 - Implemented **3D reconstruction** from camera data for grapevine/obstacle detection and path planning in unstructured environments.
 - Explored CV model optimization for **edge deployment** and real-field robustness (dataset variance, lighting conditions, occlusion handling).
 - Focus: production-oriented R&D system design for commercial implementation, not only research proof-of-concept.

Relevant Projects

- Dec 2024 – Feb 2025 **InterCropGym – RL for Agricultural Systems**, *Python, PyTorch*
- Extended CropGym with LINTUL3 engine; implemented DDQN, SAC, PPO for multi-crop optimization.
 - Applied reinforcement learning to precision agriculture decision-making systems.
- April 2025 – May 2025 **Differential Dynamic Programming (DDP)**, *Python, Control Theory*
- Implemented **trajectory optimization algorithms** for non-linear robotic systems using iterative LQR.
 - Applied DDP for optimal control problems with complex dynamics and state/control constraints.
- Aug 2024 – Sept 2024 **3D Simulation Engine**, *JavaScript, Three.js, Cannon.js*
- Built physics-based simulation with rendering, collision detection, pathfinding AI for test environments.

Technical Skills

Robotics	ROS 2, ROS 1, Computer Vision (OpenCV), Path Planning, Motion Control
AI & Vision	PyTorch, Object Detection, Edge Optimization, Real-Time Inference, NumPy
Languages	Python, C++, MATLAB, JavaScript
Simulation	Gazebo, CoppeliaSim, Physics Engines (Cannon.js), Simulink
Engineering	Fluid Dynamics, Thermal Systems, Propulsion, Control Theory, Hydraulics
Tools	Git, Docker, Linux, LaTeX

Education

- Sep 2023 – Present **MSc in AI & Robotics**, *Sapienza University of Rome*
Focus: **Autonomous Systems, Machine Learning, Robotics Control, Computer Vision**
- Jul 2018 – Jul 2023 **BSc in Aerospace Engineering**, *Politecnico di Milano*
Focus: **Fluid Dynamics, Propulsion Systems, Thermal Management, Control Systems**
Thesis Project: Soyuz rocket liquid propellant redesign (engine performance optimization, thermal analysis)

Interests & Volunteering

- Volunteering **Amare Onlus (Ethiopia)**: Documented no-profit development projects.
- Interests Sustainable Agriculture, Precision Farming, Autonomous Systems, Environmental Biology, Open Source Robotics.